Kavindra Kumar

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EDUCATION

National Institute of Technology, Jamshedpur 2023 – Present
Bachelor of Technology, Electrical Engineering CGPA: 7.72/10

B.L.I.A Public School, Aurangabad 2023
Class XII (CBSE) 88.4%

B.L.I.A Public School, Aurangabad 2021
Class X (CBSE) 85.8%

ACHIEVEMENTS

- National Semi-Finalist – Flipkart GRiD 7.0
- **DSA & Competitive Programming:** Codeforces Specialist (Max Rating: 1445) | LeetCode 1800+ | CodeChef 4-Star (1835)
- Solved **1000+ Data Structures and Algorithms problems in C++** across platforms, optimizing for strict memory and time complexity constraints.
- Secured **2nd Place among 150+ teams** at Hack Horizon 2.0 by developing CitizenBridge, a microservices document processing platform.
- Secured **3rd Place among 100+ teams** at GDG Ranchi Hackathon by developing IPL Akinator, an AI-powered cricket player guessing application.

TECHNICAL SKILLS

Languages	C/C++, SQL, JavaScript
Core Fundamentals	Data Structures & Algorithms, Object-Oriented Programming (OOP), Operating Systems, DBMS
AI & Tools	RAG (Retrieval-Augmented Generation), Git, GitHub, Postman
Web Technologies	React.js, Node.js, Express.js, MongoDB, PostgreSQL, Prisma, SQL, Tailwind CSS

PROJECTS

CodeX – Full-Stack Coding Platform GitHub *Apr 2025 – May 2025*
Node.js · Express · PostgreSQL (Prisma) · React · Monaco Editor · Zustand · Tailwind CSS · Judge0 · Docker

- **Developed a full-stack DSA platform using the PERN stack**, enabling users to solve algorithmic problems, track submission history, and create and manage custom playlists.
- **Integrated Judge0 through a dedicated backend execution service** to securely compile and evaluate multi-language code submissions with isolated execution workflows.
- **Architected modular frontend state management using Zustand** by separating authentication, code execution, and playlist logic into independent stores, improving data flow predictability and UI performance.
- **Designed a relational database architecture using Prisma ORM** to model relationships across users, problems, playlists, and submission histories.
- **Implemented secure JWT-based authentication and role-based access control** using custom middleware and protected frontend routes for authorized administrative operations.

CitizenBridge GitHub *Apr 2026*
RAG · Node.js · Express · React (Vite) · MongoDB · Qdrant · Cloudinary · Clerk · Inngest

- Built an end-to-end **RAG-based document intelligence system** for vector-based semantic question answering and automated workflow generation.
- **Architected an asynchronous, event-driven document processing pipeline using Inngest**, offloading resource-intensive parsing tasks from the main Node.js API server to enable non-blocking background processing.
- **Integrated Qdrant vector database** for document embeddings and cosine similarity-based semantic search, enabling context retrieval for RAG-powered responses.
- **Streamlined identity and document state management** using Clerk webhooks for authentication events and Cloudinary for secure file storage, with document metadata managed through MongoDB schemas.

Smart Street Light – Energy Efficient Highway Lighting *CCA Academic Project*
ESP32 · C++ · IR Sensors · LDR · MOSFETs · LEDs

- **Developed an ESP32-based embedded highway lighting system** that detects approaching vehicles using distributed IR sensors and controls individual street lights for energy-efficient operation.
- **Implemented sequential lighting control** where lights ahead of the vehicle turn ON while lights behind it turn OFF as the vehicle moves through successive sensor zones.
- **Integrated LDR-based day/night detection and GPIO control** using C++ firmware to automatically activate the required lighting section and switch all lights OFF when no vehicle is detected.

LEADERSHIP & CERTIFICATIONS

- **E-Cell NIT Jamshedpur** | Assisted in organizing entrepreneurship events, workshops, and outreach initiatives for the student community.
- **Certifications:** Postman Student Expert (API Fundamentals), Full-Stack Development Certification (GeeksforGeeks & ChaiCode)
- **SQL (Basic) Certification, HackerRank** | Validated foundational knowledge of SQL and relational database queries.